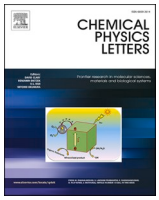
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Controlling electrospun nanofibers surface morphology using relative humidity for enhancing triboelectric nanogenerator performance

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ABSTRACT

We employ an electrospinning technique to precisely modulate nanofibers diameter by varying humidity, impacting surface morphology, roughness and frictional contact-area of triboelectric materials. This leads to an augmentation of surface charge density, promoting generation of surface charges on triboelectric materials. MXene (MX) and cellulose acetate (CA) form an effective pairing, with optimized sample CA-40_MX-40 at 40 % RH displaying five-fold increase in voltage compared to CA-20_MX-20. This sample achieves a peak power density of 2351.1 mW/m^2 at 10^6 Ω, capable of powering LEDs. These findings introduce a viable fabrication technique to manipulate surface roughness, promising significant avenue for sustainable high-performance TENGs.

1. Introduction

In the current era of "Digital Age", the rapid advancement of nanotechnology has resulted in the proliferation of highly integrated electronic devices and Internet of Things, leading to significant interest [1,2]. As a result of these developments, there is a higher demand for micro-to-milliwatt power sustainable energy sources. Traditional energy storage units, such as capacitors and batteries, are constrained by their limited capacity necessitate frequent recharging and replacement [3,4]. To address this challenge, researchers worldwide have dedicated extensive efforts to harness and utilize biomechanical energy generated during our routine activities like walking, cycling, knee joint motion, and even finger tapping [5,6]. Dated back to 2012, Wang and his team developed the first triboelectric nanogenerator (TENG), which can transform any sort of randomly distributed mechanical energy to electrical energy *via* the mutually reinforcing effects of triboelectrification and electrostatic induction [7,8].

So far, plenty of research efforts are devoted to increase the performance of TENG devices. Utilization of material selection with high surface charge density [9,10], surface modification [11] and structure optimization are key elements, which can promote the triboelectric effect and yield the high output performance. As this mechanism involves the constrained static charges available on material's surface to induce charge transfer or electric potential, thus the high surface-charge density is considered as critical parameter to define high output performance of TENG [8,11]. Some interesting works have moved in this research field of exploring the multi-scale surface roughness of triboelectric materials to achieve high output performance [12]. Numerous surface modification approaches have been employed such as pattern transferring technique, dry plasma etching, inkjet printing, chemical treatment, photolithographic mold-base, and cold compression [13,14]. For instance, Chen et al. worked on microcontroller crumpled morphology of gold films, which increased the surface roughness and triboelectric charge densities of the material, resulting in power density of 0.22 mW/cm^2 [15]. Zhen Wen et al. used blade coating method to fabricate controllable conductive and rough structured poly(3,4-ethylenedioxythiophene):poly(4-styrenesulfonate) electrode based TENG and achieved a maximum power density of 4.06 mWm^{-2} [16]. Nevertheless, above mentioned existing methods suffer various drawbacks such as, time consuming, expensive and some of them are complicate to fabricate as they required high temperature and tedious vacuum during the synthesis process.

To break through these limitations, an electrospinning approach has been proposed as a simple, inexpensive, and easy-to-use technology. As the surface structure formed by electrospinning technique imparts various properties such as ease of forming fibers from few nm to several μm, high specific surface area, good porous structure, higher surface roughness, ease of functionality, excellent mechanical properties, high flexibility and so on [17]. In recent years, many researchers have formed structure modified material for TENG application by using the

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electrospinning technique. For instance, Zhang et al. elaborated the nanoporous and rough nanostructure of PVDF nanofibers mat prepared *via* facile electrospinning technique which endow the enhanced electrical performance of corresponding TENG because of nanofibrous structure possessing high specific surface area for frictional contact and thus, high surface charge density [18]. In other study, Luo and his co-workers utilized the electrospinning technique to prepare binary blend PVDF/PS NFs. Through this approach, they have achieved high surface roughness and thus, high surface charge density for TENG [19]. Jiang et al. employed all electrospun MXene/PVA composites-based TENG and achieved a maximum output power density of 1087.6 mW/m^2 [20]. More importantly, the electrospinning technique facilitates the control on surface morphology of processed material through set up parameters such as temperature, relative humidity (RH), polymer concentration, drum rotation per minute, tip to collector distance, applied voltage, flow rate etc. [21,22]. As the spinning solution jet injected from needle experiences viscoelastic stress and whipping instability before deposition and collection of nanofibers, the parameters such as temperature and RH play a vital role to control the surface roughness of processed nanofibers (NFs) [23,25].

With perspective to material selection, choosing the materials that are far apart on the triboelectric series tends to enhance the triboelectric effect, resulting in greater electrical output [11]. In this perspective, a class of two-dimensional (2D) transition metal carbonitrides, MXenes $Ti_3C_2T_x$ [25], have been widely acknowledged as potential candidate due to their prominent properties such as high specific area, high metallic conductivity, and outstanding mechanical as well as electrical properties [26]. Additionally, the abundance of surface functional groups –F and terminating functional group containing oxygen on MXene results in an excellent electronegativity, making it a promising alternative to electronegative materials for TENGs [27,28]. In the recent years, MXene composite nanofibers owing to their fascinating features such as light weight, flexibility, high surface-to-volume ratio, high mechanical strength are flourishing the field applications such as high-performance energy harvesting and self-powered sensing [29–33]. On the other hand, cellulose being a natural abundant organic material, possesses biodegradable and biocompatible properties along with a robust electron donating capacity [34,35]. Due to these characteristics, cellulose acetate (CA) based NFs is considered as tribopositive layer in the context of TENGs [36]. In the recent years, a rapid progress has been noticed based on MXene/PVA NFs paired with CA NFs for developing high performance TENG. However, keeping the same combination of materials, the surface modification of the NFs to be further tuned in terms of contact-electrification process to maximize the frictional contact-area. This will provide a way to systematically investigate the role of surface roughness on the electrical performance of TENGs.

Considering the discussion, in the present work, we introduced all-electrospun TENG device by pairing MXene/PVA NFs with CA NFs. The surface roughness of prepared NFs was controlled by altering the processing RH conditions (20 % – 60 % RH) while keeping all other parameters constant. The varying RH values significantly influenced the NFs diameters and hence, the NFs packing density, which in turn led to change in surface roughness. AFM studies revealed that MXene/PVA and CA NFs processed at 40 % RH exhibited higher surface roughness and thereby, outperformed with larger output voltage of ~ 140 V and instantaneous peak power density of ~ 2351 mW/m^2. The integrated TENG also showed the good cyclic stability test over 3000 s. Further, the TENG capability practical applications was demonstrated by powering of LEDs. Therefore, the findings of present study provide an effective avenue for the performance enhancement of TENG.

2. Experimental section

2.1. Materials

Pristine Ti_3AlC_2 powder (MAX) was obtained from Nanoshel UK Ltd. Cellulose acetate (CA) (M_w ~ 30,000), polyvinyl alcohol (PVA) (M_w ~ 86,000), hydrofluoric acid (HF), ethanol, acetone, N, N-dimethylformamide (DMF) were purchased from Merck, India. All the materials were utilized directly as such.

2.2. Synthesis of 2D MXene $Ti_3C_2T_x$

2D MXene was prepared through etching of parent MAX powder. In typical synthesis, firstly, 2 g of MAX powder was gradually added to the 40 mL HF solution while being continuously stirred for 24 h. The resultant solution was then centrifuged (@3500 rpm; 10 min per cycle) and rinsed with DI water until the supernatant reached a pH value of 7. Finally, the MXene was obtained by drying the supernatant solution in oven at 60 °C for 24 h.

2.3. Preparation of CA and MXene NFs

CA and MXene NFs were prepared using electrospinning technique (Fig. 1). To process the NFs in different RH levels (20 %, 30 %, 40 %, 50 % and 60 %), a humidifier was used in electrospinning chamber. CA NFs were prepared by blending polymeric solution containing 19 wt% of CA powder in 15 mL solution of DMF/acetone (1:2 v/v). The solution was then transferred into 5 mL syringe and extruded through a needle with diameter of 13.2 mm. The syringe pump flow-rate was set at 0.5 μL/s while the 17 kV high voltage was applied between the metallic needle and the ground collector at a tip-to-collector distance of 15 cm. Other side, MXene NFs were prepared by blending 10 wt% PVA solution into 9 mL DI water followed by mixing 30 wt% MXene solution. The spinning solution parameters were fixed as flow-rate of 0.2 μL/s, high voltage of 24 kV and needle-to-collector distance of 15 cm.

2.4. Fabrication of TENG and electrical measurements

The prepared electrospun CA NFs and MXene NFs mats with dimensions of 3 cm × 3 cm were paired as positive and negative triboelectric materials to fabricate TENG. The aluminum sheets on which the NFs were collected used as substrate as well as flexible electrodes for TENG. Further, the copper wires were connected at the back side on the aluminum sheets to complete the TENG circuit. The fabricated TENG was operated using home-made tapping device and its output performance was measured using Keysight DSO1052B digital storage oscilloscope (Fig. S1). Further, TENG electrical characteristics were tested under different frequencies using frequency modulator and under different external load resistors.

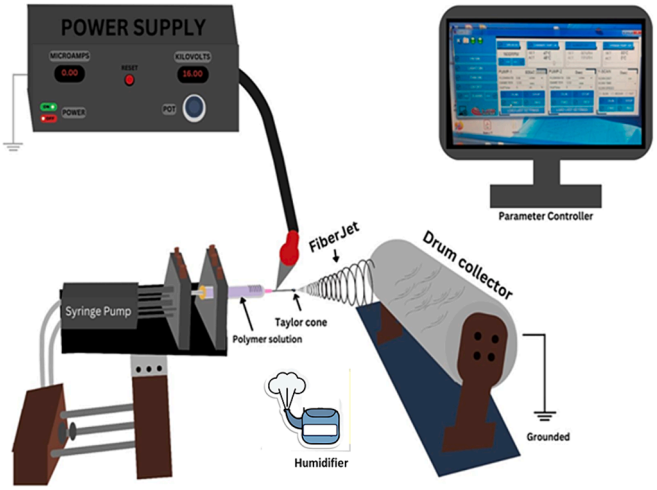


Fig. 1. Schematic view of electrospinning setup.

2.5. Characterization tools

The structural properties of prepared samples were investigated using X-ray diffraction instrument (XRD; Bruker D8 Focus) using Cu $K\alpha_1$ irradiation source ($\lambda = 1.5406$ Å) in the range of 2θ ($5 - 70^\circ$). The surface morphology of the electrospun nanofibers was investigated using field emission scanning electron microscopy (FESEM; Carl Zeiss, Supra 55). The surface roughness of electrospun nanofibers was evaluated by atomic force microscopy (AFM; Nanosurf Naio AFM atomic force microscope system).

3. Results and discussion

3.1. Material characterization

To analyze the crystallographic information and structural formation of the fabricated samples, XRD was conducted. Fig. 2 (a) shows the comparative XRD pattern of bulk MAX precursor and synthesized MXene. MAX precursor displays the prominent diffraction peaks at 9.64°, 19.24°, 39.02°, 41.92° and 60.28° corresponds to (0 0 2), (0 0 4), (1 0 4), (1 0 5) and (1 1 0) characteristic crystallographic planes, respectively [37]. In MXene XRD pattern, the disappearance of sharp intense peak at 39.02° suggests the removal of Al and thus, the successful structural transformation of parent MAX to MXene [38]. As is known, there is slight shifting of (0 0 2) and (0 0 4) planes towards lower angles from 9.64° to 8.66° and 19.24°to 17.94° which suggests structural expansion of interplanar spacing of MXene which is due to intercalation of water molecules and functionalization of surface groups (−O/OH, −F) during the etching process [39]. Further, the prepared electrospun CA NFs and MXene/PVA NFs were characterized. Fig. 2 (b) illustrates that the XRD pattern of CA NFs exhibits the broad hump at about 21.9° which depicts the semi-crystalline nature of the prepared nanofibers. In XRD pattern of MXene/PVA NFs, the strong diffraction peak at 19.14° corresponds to (1 0 1) plane of PVA polymer molecules. The insertion of PVA molecules causes the increase in interplanar spacing of MXene as the (0 0 2) peak of MXene significantly shifted from $2\theta = 8.66^\circ$ to 7.36° which is due to the deposition of PVA molecules which increases the distance between the MXene nanosheets.

Fig. 3 shows FESEM images of CA NFs and MXene/PVA NFs that were fabricated under the different RH conditions specified by controlled electrospinning parameters. The prepared NFs are coded as CA-RH and MX-RH, where CA stands for cellulose acetate, MX for MXene and RH for varying humidity conditions i.e., 20 %, 30 %, 40 %, 50 % and 60 %. The morphology of prepared NFs is found to be varied with respect to RH. The distribution of nanofibers diameter data fitted by Gaussian function is displayed in histograms to determine average nanofibers diameters. The results displayed the average diameters of CA NFs and MXene/PVA NFs are in increasing trend with increasing RH processing conditions (Table 1).

These effects can be attributed to changing viscoelastic properties and Rayleigh jet stability of polymeric solution injected from needle tip in the varying RH% conditions. At low RH%, the water vapors are present in lower content which evaporate rapidly and thus, polymeric jet solution undergoes rapid solidification. Therefore, the applied electrospinning voltage compensates the surface tension of jet solution for higher stretching between needle-tip and collector and thus, leads to formation of NFs with thinner diameters. The increase in RH% processing conditions result in the formation of wet NFs because of large amount of water ingress in polymer jet causing them to fuse together before evaporating and thus, lead to thicker NFs [40–42] observed in Table 1. Notably, at RH conditions of 50 % and 60 %, the beaded NFs were formed; this is because the jet solution became too viscous, causing Rayleigh jet instability and thus, leading to coagulation of water vapors in the form of beads with decrease in NFs diameter [24]. Moreover, the much lower and higher RH conditions could not allow to process the NFs; this is because the extremely lower RH values result in large viscoelastic stress due to negligible water ingress in polymeric solution which is high enough to compensate by applied voltage, while the extremely higher RH values raise the instability in jet formation. This is why, in the present work, CA NFs were not processed below 20 % RH and above 60 % RH, while for MXene/PVA NFs, 20 % RH and 50 % RH were the saturation processing conditions.

Further, the high-resolution SEM images of CA-40 and MX-40 NFs were shown (Fig. S2a-b), where the formation of small pores can be seen on the surface of NFs. This could contribute to enhanced surface roughness of NFs and hence, the higher surface charge density of triboelectric layer. Furthermore, TEM studies were conducted to elucidate the formation of CA-40 and MX-40 NFs (Fig. S2c-d). From high resolution TEM images (Fig. S2e-f), the interplanar spacing of CA-40 and MX-40 NFs was found to 0.40 nm and 0.29 nm corresponding to their respective ($0\bar{1}10$) and (2 0 0) planes.

The other more interesting effect caused by varying RH is the surface roughness. AFM was conducted to study the influence of RH on the surface roughness of CA NFs and MXene/PVA NFs. The surface roughness line profile of all the samples were shown in Fig. 4. The value of root-mean square surface roughness (R_q) of CA NFs increases in a

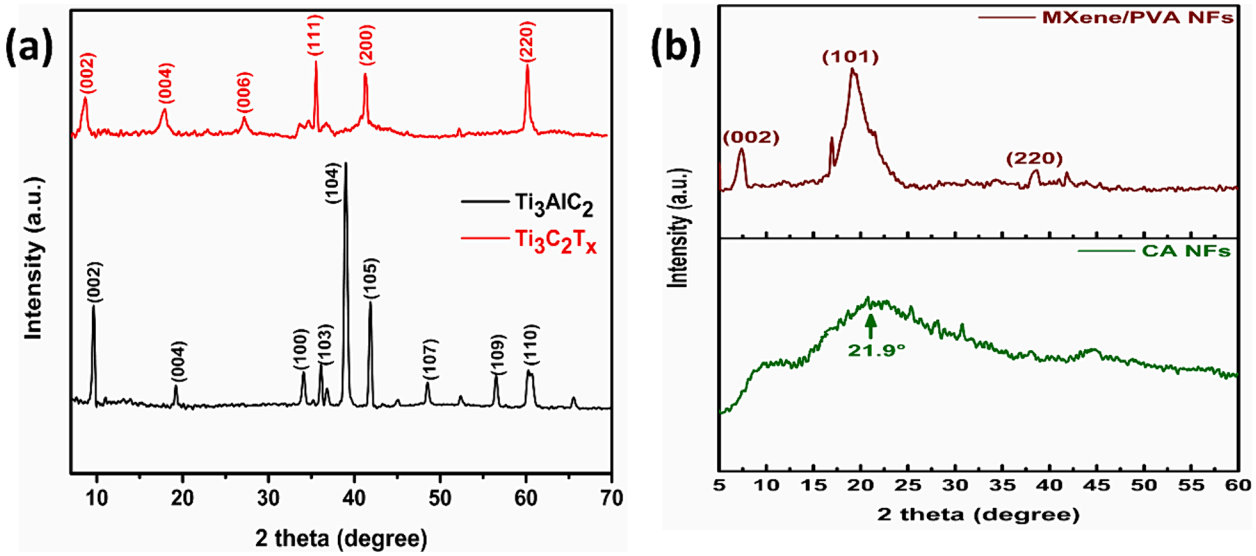


Fig. 2. XRD diffractogram of **(a)** MAX phase and prepared MXene, **(b)** synthesized CA NFs and MXene/PVA NFs.

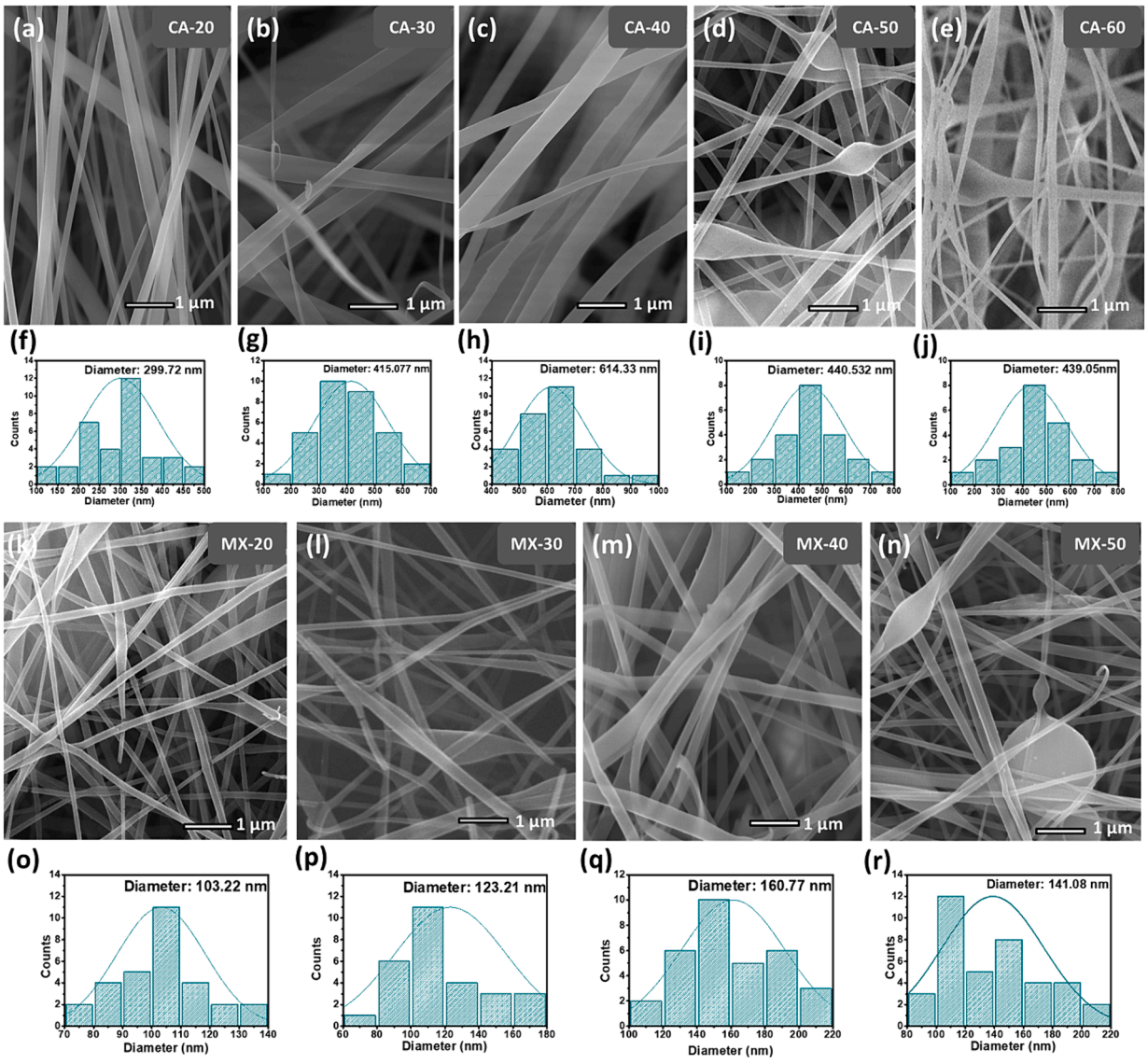


Fig. 3. **(a-e)** FESEM images and **(f-j)** fibers diameter histogram analysis of CA-20, 30, 40, 50 and 60 fibrous membranes and **(k-n)** FESEM images and **(o-r)** fibers diameter histogram analysis of MX-20, 30, 40 and 50 fibrous membranes.

Table 1
Variance of the measured average diameter of CA and MXene/PVA electrospun NFs.

Relative humidity (RH)	Diameter of CA NFs	Diameter of MXene/PVA NFs
20 %	299.72 nm	103.22 nm
30 %	415.077 nm	123.21 nm
40 %	614.33 nm	160.77 nm
50 %	440.532 nm	141.08 nm
60 %	439.05 nm	–

manner of 3.6370 nm < 4.6598 nm < 9.8388 nm corresponding to C-20, C-30, and C-40 NFs, respectively, and then decreases as 7.1022 nm and 4.3220 nm for C-50 and C-60, respectively. Similarly, for MXene/PVA NFs, the R_q value increases as 4.3255 nm < 4.5482 nm < 12.0500 nm for M−20, M−30 and M−40, respectively and then decreases to 9.8809 nm for M−50. The correlation of increasing surface roughness with increasing RH% processing conditions can be attributed to reduced packing density of NFs, which is in conjunction with increasing NFs diameters[30]. As observed from FESEM images, the rise in RH processing conditions i.e., 20–40 % RH decreased the packing density of NFs, which in turn increased the pore-volume of NFs and thus, increased the surface roughness. However, the R_q value decline for higher RH processing conditions may be accounted to formation of beaded NFs, which resulted in relatively increased packing density.

3.2. *Functional mechanism and electrical characteristics of TENG*

To evaluate the electrical characteristics of TENG, the device was fabricated (3 × 3 cm^2), consisting of two layers, one is triboelectric positive layer: CA-NFs, and other is triboelectric negative layer: MXene/PVA NFs. A self-made tapping device as shown in Fig. 5(a) has been used for measuring the electrical properties. Fig. 5(b) depicts a schematic representation of functional mode of TENG in terms of device operation i.e., vertical contact-separation mode. In initial state, the layers are separated from each other and there is no surface frictional-contact between them and hence, no triboelectric charges are induced on surface. When external force is applied, the gap between the layer's decreases, bringing two surfaces of different charge densities in contact and hence, causing the transfer of charge. The ability to gain electrons is significantly high for MXene, therefore, the net negative charge is induced on MXene-NFs and net positive charge is induced on CA-NFs.

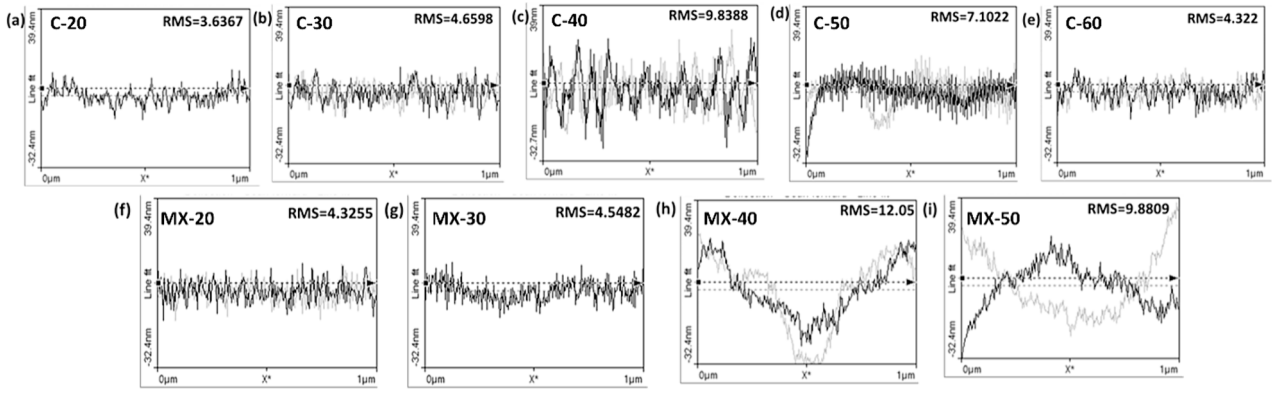


Fig. 4. (a-e) Surface roughness profiles of synthesized CA-20, 30, 40, 50 and 60 electrospun fibrous membranes and **(f-i)** MX-20, 30, 40 and 50 electrospun fibrous membranes.

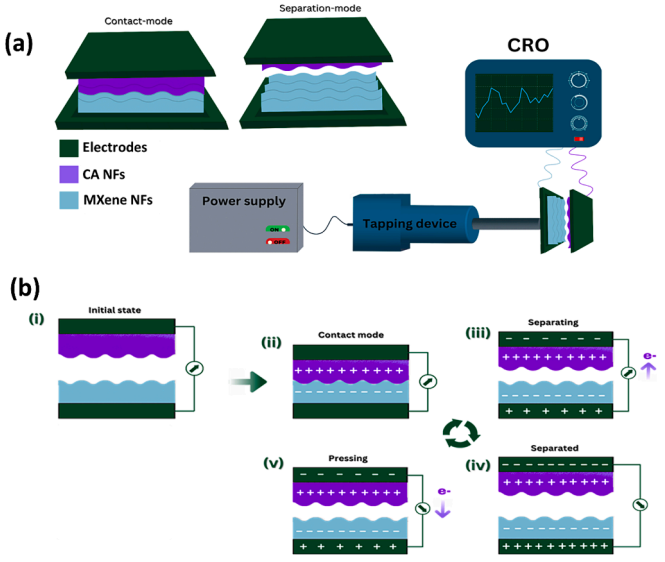


Fig. 5. Schematic illustration of **(a)** home-made tapping device **(b)** principle of vertical contact separation mode of TENG device and flow of positive/negative charges.

On releasing the force, the gap between layers increases and the electric field is generated, causing the free electrons to flow between external electrodes. Once the layers are separated, the generated electric field gets neutralized. On applying external force again, the process of electron transfer and electric potential generation takes place in opposite direction to complete the working cycle. This way, TENG produces an alternating potential on repetition of contact-separation cycles.

Further, an array of 16 combinations of fabricated CA and MXene/PVA NFs synthesized under different relative humidity conditions were made and tested comprehensively. The corresponding output waveforms in the term of voltage are shown in Fig. 6(a-e). and listed in Table 2. Referring to TENG consists of CA-NFs (20 RH %), the output voltage is ~ 14 V for CA-20_MX-20 pairing, which increases to ~ 24 V when it is paired with MXene NFs (40 RH %) i.e., for CA-20_MX-40. Further, the output voltage increases to ~ 28 V for CA-30_MX-40 and finally, it reaches to maximum value of ~ 69 V for CA-40_MX-40 pairing. The reason for the significant voltage enhancement is likely due to the increasing frictional contact area. As the contact area increases more will be the active sites on the surface, facilitating an efficient transfer of electrons between the triboelectric layers through the triboelectric effect [42]. The contact area is likely dependent on surface-friction properties of nanofibers, which found to be maximum for 40 % RH as analyzed in AFM studies. However, the output voltage drops off for the TENG paired with nanofibers prepared at RH of 50 % and 60 %. The fundamental reason of the reduction in output voltage of the TENG device is the

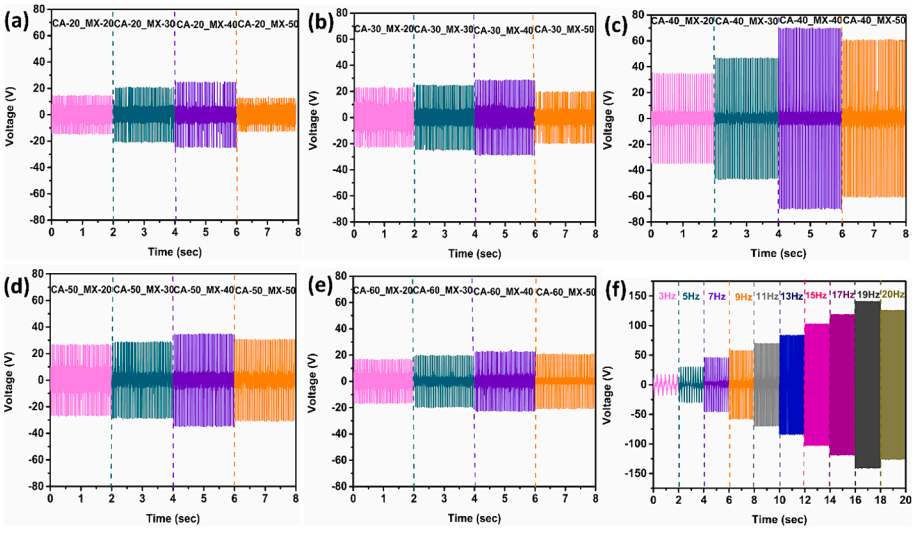


Fig. 6. (a-e) Triboelectric output voltage corresponding to different pairing of NFs prepared under different RH condition, **(f)**Frequency dependency of TENG output voltage.

Table 2
Output voltage corresponding to various combinations of CA and MXene/PVA NFs.

Cellulose acetate (CA)	MXene (MX)	TENG samples	Open-circuit voltage
CA @20 % RH	MX @20 % RH	CA-20_MX-20	14 V
	MX @30 % RH	CA-20_MX-30	20 V
	MX @40 % RH	CA-20_MX-40	24 V
	MX @50 % RH	CA-20_MX-50	12 V
CA @30 % RH	MX @20 % RH	CA-30_MX-20	22 V
	MX @30 % RH	CA-30_MX-30	24 V
	MX @40 % R H	CA-30_MX-40	28 V
	MX @50 % RH	CA-30_MX-50	19 V
CA @40 % RH	MX @20 % RH	CA-40_MX-20	34 V
	MX @30 % RH	CA-40_MX-30	46 V
	MX @40 % RH	**CA-40_MX-40**	**69 V**
	MX @50 % RH	CA-40_MX-50	60 V
CA @50 % RH	MX @20 % RH	CA-50_MX-20	26 V
	MX @30 % RH	CA-50_MX-30	28 V
	MX @40 % RH	CA-50_MX-40	34 V
	MX @50 %RH	CA-50_MX-50	30 V
CA @60 % RH	MX @20 % RH	CA-60_MX-20	16 V
	MX @30 % RH	CA-60_MX-30	19 V
	MX @40 % RH	CA-60_MX-40	22 V
	MX @50 % RH	CA-60_MX-50	20 V

negative influence of high humidity, which diminishes the impact of the triboelectric material by reducing the surface roughness of active material.

The theoretical studies have revealed that the thickness of sample influence the surface charge-density and hence, the output TENG performance [43]. For this study, the nanofibrous mats of CA-40_MX-40 of different thickness (0.01–0.07 nm) have been prepared by varying the processing time of electrospinning. In the present work, the TENG performance has found to increase with increasing thickness of nanofibrous mats and reached at highest voltage of 69 V for mat thickness of 0.05 mm (Fig. S3). The lower thickness (<0.05 mm) of nanofibrous mat results in lower output voltage due to the non-uniform and limited distribution of surface charges. Other side, the thicker nanofibrous mats are supposed to have lower-surface area, depriving the surface charge-density and hence, the output voltage (Fig. S3) [44]. Additionally, the impact of tapping frequency was noted on TENG output voltage. Fig. 6 (e) shows the open-circuit voltage of CA-40_MX-40 sample steadily rises from 17 V to 140 V with increasing frequency from 3 Hz to 19 Hz. Increase in frequency causes the CA NFs and MXene NFs to make contact more quickly, which increases the amount of charge transfer across the layers in shorter period and hence, the larger output voltage. The decrease in voltage is observed for further increase in frequency i.e., at 19 Hz. This is because at higher frequencies, the separation process occurs too frequently such that the time of contact between layers decreases, causing inappropriate charge transfer between the layers. Further, TENG performance is directly correlated with applied force, which follows that the stronger tapping force between the layers leads to closer or larger frictional contact, contributing to more charge-transfer and in turn, the higher TENG output voltage [45]. In the present work, the output voltage shows increasing trend with increase in external force and achieved a maximum voltage of 69 V at 8 N and after that no significant change observed in the output voltage (Fig. S4). The reason for this saturation in output voltage is due to less significant change in real contact-area between the layers [20]. Even the much stronger tapping forces result in the degradation of TENG structure, which could also be reason for depriving output voltage enhancement.

To investigate the functionality of fabricated TENG for practical applications, a long-term reliability test was performed for 3000 cycles under similar atmospheric conditions. Fig. 7(a) shows the open-circuit voltage remained uniform, demonstrating the excellent reliability and stability of the device for this work. Further, the output performance of CA-40_MX-40 pairing-based TENG was evaluated under different external load conditions ranging from 10^1 to $10^9\Omega$. Fig. 7(b) illustrates that the voltage increases with increase in external load and the current follows the reverse trend in accordance with ohm's law, achieving the maximum output voltage of 148 V and current of 110 μA. Furthermore, the fitted instantaneous power density according to equation, $P = \frac{V^2}{R}$, gives the maximum value of ~ 2351.1 mW/m^2 at 10^6 Ω as shown in Fig. 7(c).

For comparison of performance, CA_MX film-based TENG was fabricated, showing the output voltage of 8 V when operated at optimized 19 Hz tapping frequency (Fig. S5a). Under different external loads (10^1 –10^9 Ω), the fabricated TENG achieved the maximum open-circuit voltage of 14 V and short-circuit current of 22 mA with maximum instantaneous power density of 84.1 mW/m^2 at 10^6 Ω (Fig. S5b-c). In comparison, the triboelectric performance of CA_MX NFs-based electrode was found to higher owing to the higher surface charge density caused by large surface-to-volume ratio of one-dimensional structure of processed NFs. The other advantage of using NFs-based electrode is high recyclability, which was confirmed through SEM and TEM investigations. After cycling test, SEM images (Fig. S6a-b) of CA-40 and MX-40 showed no noticeable changes after cycling tests, where the nanofibrous morphology is well maintained. Similarly, TEM images (Fig. S6c-d) of CA-40_MX-40 after cycling tests showed no significant change, confirming the stability and recyclability of prepared nanofibrous films.

As summarized in Table 3, we further compared the output characteristics of electrospun TENGs based on previous literature with the present work. Our findings revealed that the TENG using CA paired with MXene/PVA NFs fabricated under varying humid processing conditions exhibited superior performance in terms of output voltage and power density when compared to previously reported TENG devices. It is worth highlighting those prior studies predominantly employed NFs prepared under ambient conditions resulting in relatively flat and smooth fibrous surfaces. For instance, Ping et al. fabricated MXene/PVA nanofibers based TENG using electrospinning technique at ambient humidity and reported maximum instantaneous power density of 1087.6 mW/m^2

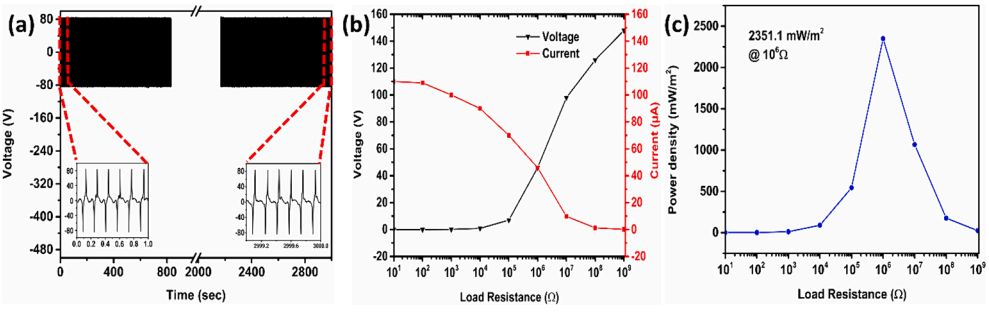


Fig. 7. Electrical performance of TENG **(a)**Cyclic test of durability and stability for 3000 s, **(b)**Dependence of output voltage and current on external loading resistance and **(c)** Power density as the function of loading resistance.

Table 3
Compare the performance of the Triboelectric Nanogenerator as per our latest findings against previously published research.

Material	Maximum Output voltage	Maximum power density	Reference
MXene/PET-ITO	~500 to ~ 650 V	~0.5 – 0.65 mW/m^2	[28]
Silk-PVA/MXene	118.4 V	~1087.6 mW/m^2	[20]
PVA/Ag-fluorinated ethylene propylene	530 V	359 mW/m^2	[46]
CA-MXene/PVA	~100 V	1361 mW/m^2	[47]
PVDF-Cellulose	106.2 V	13.3 W/m^2	[48]
CA-PDMS	25 V	16 mW/m^2	[49]
A-MXene/PVA-A-CA	54 V	1102 mW/m^2	[50]
CA-MXene	140 V	2351.1 mW/m^2	This study

[20]. In the present work, the humidity processing conditions (20–60 % RH) of electrospinning chamber have been varied and their influence on the fabrication of MXene/PVA nanofibers has been studied. The correlation between the humidity conditions and the diameter of processed nanofibers has been observed, which in turn, found to influence the packing density and surface roughness of nanofibers. The results showed that MXene/PVA nanofibers processed at 40 % RH exhibit larger diameter of 160.77 nm and higher surface roughness of 12.05 nm, resulting in enhanced output performance of TENG with maximum instantaneous power density of 2351.1 mW/m^2. Further, to test the reproducibility of results, the five samples of CA-40 and MX-40 each have been synthesized and studied under similar experimental conditions; showing no significant deviation in output TENG voltage (Fig. S7). Furthermore, the TENG output performance has been tested with respect to environmental humidity (44–84 % RH). The open-circuit voltage of CA-40_MX-40 based TENG shows decreasing trend with increase in humidity level because of dissipation of surface charges due to adsorption of water molecules on the hydrophilic surface of MXene. The voltage found to decrease from 69 V to 30 V under maximum relative air moisture of 84 % (Fig. S8).

Furthermore, the TENG device is directly connected to the full wave bridge rectifier which is responsible for converting alternating current output signals into direct current output signals. The output power generated from this device is utilized to simultaneously drive 24 commercial LEDs forming the well-recognized character “GNDU” as illustrated in Fig. 8. This indicates that the output voltage of these TENG materials make it possible to power specific wearable electronic devices, showcasing their significant potential for practical use.

4. Conclusion

In summary, this study demonstrated a facile and controlled method for modifying the surface morphology of the triboelectric materials which has been regarded as the most efficient approach to improve the electrical performance of TENGs. Utilizing the electrospinning approach, we developed CA and MXene nanofibers (NFs) with different diameters by varying relative humidity (RH) processing conditions from 20 % to 60 % RH. FESEM images revealed the effect of RH on NFs morphology, showing the variations in NFs diameter and thus, NFs packing density. AFM showed a correlation between surface roughness and RH processing conditions. The output voltage of the TENG has been significantly influenced by the pairing of NFs synthesized under different RH conditions. The maximum voltage output has been observed for CA-40_MX-40 pairing, attributed to the increased contact area and enhanced surface roughness. Furthermore, the tapping frequency exhibited a direct impact on the device’s performance, with an optimal output observed at 19 Hz which is 148 V. The device has also demonstrated excellent reliability and stability during a 3000-cycle durability test. Additionally, the TENG’s practical applicability demonstrated by connecting it to a full wave bridge rectifier and using the generated power to illuminate a set of commercial LEDs. This underscores the potential of the TENG to power wearable electronic devices, highlighting its significance in real-world applications.

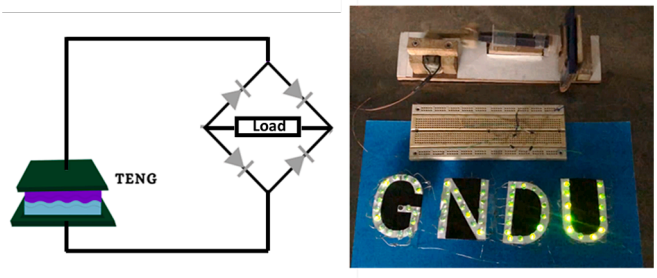


Fig. 8. Schematic circuit for TENG connection and Lightning of LED.

CRediT authorship contribution statement

Ishu: . Sagar Sardana: Writing – review & editing, Methodology, Conceptualization. **Aman Mahajan:** Writing – review & editing, Visualization, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cplett.2024.141289.

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